Fountain View Hall

Release 0.2.0

Derek R. Neilson

Jun 10, 2026

CONTENTS:

Fountain View Hall's management tool, `fvh-manager` is used to manage reservations at Fountain View Hall.

CHAPTER

ONE

USER GUIDE

This guide will walk you through `fvh-manager` and covers topics such as how to install and use fvh-manager.

1.1 Installation

1.1.1 User Prerequisites

To verify you meet all prerequisites, run:

```
uv --version
```

Expected output:

```
uv 0.x.x
```

Not:

```
command not found: uv
```

Important

Make sure the `uv` command runs before running the program.

If `uv --version` fails, use this guide to install uv.

1.1.2 Minimal Getting Started

Make sure you run these commands from the project directory

Run this once:

```
uv sync
```

Then run this to start on every subsequent run:

```
uv run fvh-manager
```

1.1.3 Activate Completion by Operating System and Shell

If desired, you can activate shell completions by running one of the following commands, depending on your operating system and shell:

Operating system	Shell	Activation command
Linux/macOS	Bash	`source completions/fvh-manager.bash`
Linux/macOS	Zsh	`source completions/_fvh-manager`
Linux/macOS	Fish	`source completions/fvh-manager.fish`
Windows	PowerShell	`. .\completions\fvh-manager.ps1`

Once shell completions are activated, make sure your virtual environment is active. You can then run the program with:

```
fvh-manager
```

To view available completions, type:

```
fvh-manager <tab>
```

1.2 Developer Installation

Meet all previous *prerequisites* and additional ones as specified bellow.

Install developer dependencies for python:

```
uv sync --all-groups
```

Verify that `pdflatex` is installed:

```
pdflatex -v
```

Verify that `git` is installed:

```
git -v
```

Verify that `lefthook` is installed:

```
lefthook -v
```

CHAPTER

TWO

REFERENCES

Command-line entry point for the project.

author
Derek R. Neilson

This module defines the project's Typer application and should remain lightweight by design.

Typer is a command-line interface (CLI) framework used throughout this project. It provides decorators such as `typer.Typer.command()` for registering CLI commands.

See also

Typer documentation

`src.main.`**`create_engine`**`(`*in_memory*`)`

Create a SQLite database engine.

Creates either an in-memory SQLite database or, by default, a persistent database at `DB_PATH`, which points to `database/data.db`.

Parameters
`in_memory` (`bool`) – If `True`, create an in-memory database. If `False`, create a persistent database at `DB_PATH`.

Returns
A SQLAlchemy engine representing the configured database connection.

Return type
`Engine`

The SQLAlchemy database URL has the general form:

```
dialect[+driver]://user:password@host/dbname[?key=value...]
```

The default project database URL has the form:

```
sqlite+pysqlite:///database/data.db
```

If `in_memory` is `True`, the database URL has the form:

```
sqlite+pysqlite:///:memory:
```

See also

SQLAlchemy Engine documentation Typer documentation

`src.main.`**`start_app`**`(`*in_memory=False*`)`

Start the CLI app.

Parameters
`in_memory` (`bool`) – If `True`, use an in-memory SQLite database.

Return type
`None`

Sets up logging and metrics for the application.

author
Derek R. Neilson

var logger
Application logger configured for OpenTelemetry.

var meter
OpenTelemetry meter used to create counters and histograms.

This module configures OpenTelemetry logging and metrics, then exposes the `track_command` decorator for tracking CLI command execution.

`src.observability.`**`configure_logging`**`()`

Configures and returns the app logger.

Return type
`Logger`

`src.observability.`**`configure_metrics`**`()`

Configures the metrics

`src.observability.`**`track_command`**`(`*command_name*`)`

Track command execution with logs and metrics.

The wrapped function records the number of command runs, active operations, errors, and command duration.

Parameters
`command_name` (`str`) – Name of the command being tracked. Format not required but should be name-of-command

Return type
`Callable`[[`TypeVar`(`F`, bound= `Callable`[`...`, `Any`])], `TypeVar`(`F`, bound= `Callable`[`...`, `Any`])]

Example

```
from src.observability import logger, track_command

@track_command("some-command")
def some_command():
    logger.info("Command executed")
```

CHAPTER

THREE

DESIGN

The design process has been documented quite thoroughly, but not all decisions are documented.

3.1 Fountain View Hall Pseudocode and User Journeys

3.1.1 Problem Statement

The project must meet all *requirements*. Besides those requirements, I have set some personal goals as well:

1. Achieve 100% test coverage.

   I have done expensive research[1] on this and have concluded that it does not guarantee code standards, but it also does not hurt code quality and is probably to make you think more about your code.

   > **Limitation of Testing**
   >
   > "Program testing can be used to show the presence of bugs, but never to show their absence."
   >
   > – Edsger W. Dijkstra [Dijkstra1970]

3.2 Overview for Fountain View Hall

3.2.1 Overview

Fountain View Hall wants a program that tracks event attendance and projected revenue for multiple bookings.

3.2.1.1 Program Requirements

- Repeatedly prompt for event name, guest count, and estimated revenue
- Continue until the user enters "done"
- Track total events processed
- Track total guests
- Track total projected revenue
- Calculate the average guests per event
- Determine the largest event

[1] See *Software Testing and Code Coverage* for a list of sources viewed to come to this conclusion.

3.2.1.2 Business Rules

- Pricing Rules
  - Large events receive a discount
- Operational Rules
  - Every booking must include a guest count
  - Booking total must be calculated before the discount is applied
  - Revenue totals should be tracked

3.2.2 Planned Technology Usage

I intended to use various technologies throughout this project; some are very necessary, and others save time. Each technology will be categorized and described below:

3.2.2.1 Editing/Developer Tools

- *Git* and *GitHub* as a VCS (version control system).[1]
- uv for package, project, and environment management.
- *lefthook* as a Git hook manager.
- *pre-commit* to run pre-commit scripts.[2]
  - *ruff* for Python linting and formatting running as a pre-commit script.
- *Sphinx* to help build this beautiful documentation.
- *pytest* as a testing suite.
- *Open Telemetry Protocol* for observability and log management.[4]

3.2.2.2 Python Packages

- *typer* for command line interface completions and argument parsing.
- *SQLite3* as a database[3]
- *SQLAlchemy* for object-relational mapping (ORM)

3.3 Sources

A curated source list for reStructuredText, OpenTelemetry, SQLAlchemy, and Python tooling.

[1] Go to the project repo to see commit history.
[2] All scripts (in the scripts directory) are written for Unix systems and use the bash language.
[4] You can visit their web page for more details, but essentially it is a web interface that helps you view logs and trends in data. Probably unnecessary, but I was eager to learn how to use this.
[3] See SQLite's home page for more details.

3.3.1 Quick Reference

3.3.2 reStructuredText and Sphinx

#	Source	Creator	Date	Link
1	**reStructuredText Markup Specification**	Goodger, David	2026	Open source
2	**Restructuredtext Primer - Material for Sphinx**	Sheppard, Kevin	8 June 2026	Open source
3	**reStructuredText Primer - Sphinx Documentation**	Sphinx Documentation	8 June 2026	Open source
4	**Welcome to the reStructuredText Primer!**	reStructuredText Primer 0.0.1 Documentation	8 June 2026	Open source

3.3.3 OpenTelemetry and Observability

#	Source	Creator	Date	Link
5	**How to Set Up a Complete Observability Stack with OpenTelemetry**	OneUptime | One Complete Observability Platform	6 Feb. 2026	Open source
6	**OpenTelemetry**	OpenTelemetry	8 June 2026	Open source
7	**open-telemetry/opentelemetry-proto**	OpenTelemetry - CNCF	22 May 2019	Open source
8	**Python OpenTelemetry**	OpenTelemetry	8 June 2026	Open source
9	**Prometheus Docs**	The Linux Foundation	9 June 2026	Open source

3.3.4 Python Libraries and Tooling

#	Source	Creator	Date	Link
10	**SQLAlchemy**	SQLAlchemy	8 June 2026	Open source
11	**SQLAlchemy ORM - SQLAlchemy 2.0 Documentation**	SQLAlchemy Documentation	8 June 2026	Open source
12	**Typer**	Typer	8 June 2026	Open source
13	**SQLAlchemy Crash Course - Master Databases in Python**	Tim Ruscica	Mar 17, 2025	Open source

3.3.5 Software Testing and Code Coverage

#	Source	Creator	Date	Link
14	**100% Test Coverage is a LIE, Here's Why…**	Sam Newman	8 June 2026	Open source
15	**Why 100% Test Coverage Isn't the Goal**	Daniel Mauno Pettersson	March 4, 2026	Open source
16	**Test Coverage**	Fowler, Martin	17 Apr. 2012	Open source
17	**Coverage Is Not Strongly Correlated with Test Suite Effectiveness**	Inozemtseva, Laura; Holmes, Reid	2014	Open source
18	**Code Coverage and Postrelease Defects: A Large-Scale Study on Open Source Projects**	Kochhar, Pavneet Singh; Lo, David; Lawall, Julia; Nagappan, Nachiappan	2017	Open source
19	**A Brief Survey on Oracle-based Test Adequacy Metrics**	Hossain, Soneya Binta; Dwyer, Matthew B.	2022	Open source
20	**Four Common Types of Code Coverage**	web.dev	2024	Open source
21	**Getting 100% Code Coverage Doesn't Eliminate Bugs**	Kapelonis, Kostis	2017	Open source

3.3.6 Detailed Source Cards

1. reStructuredText Markup Specification

Type
Specification

Author
Goodger, David

Organization
Docutils

Date
2026

URL
https://docutils.sourceforge.io/docs/ref/rst/restructuredtext.html

2. How to Set Up a Complete Observability Stack with OpenTelemetry

Type
Article

Organization
OneUptime | One Complete Observability Platform

Date
6 Feb. 2026

URL

https://oneuptime.com/blog/post/2026-02-06-complete-observability-stack-opentelemetry-open-source/view

3. Restructuredtext Primer - Material for Sphinx

Type
Documentation

Author
Sheppard, Kevin

Accessed
8 June 2026

URL
https://bashtage.github.io/sphinx-material/basics.html

4. OpenTelemetry

Type
Website

Organization
OpenTelemetry

Accessed
8 June 2026

URL
https://opentelemetry.io/

5. open-telemetry/opentelemetry-proto

Type
GitHub Repository

Organization
OpenTelemetry - CNCF

Date
22 May 2019

Accessed
8 June 2026

URL
https://github.com/open-telemetry/opentelemetry-proto

Note
Makefile referenced.

6. Python OpenTelemetry

Type
Documentation

Organization
OpenTelemetry

Accessed
8 June 2026

URL
https://opentelemetry.io/docs/languages/python/

7. reStructuredText Primer - Sphinx Documentation

Type
Documentation

Organization
Sphinx Documentation

Accessed
8 June 2026

URL
https://www.sphinx-doc.org/en/master/usage/restructuredtext/basics.html

8. SQLAlchemy

Type
Website

Organization
SQLAlchemy

Accessed
8 June 2026

URL
https://www.sqlalchemy.org

9. SQLAlchemy ORM - SQLAlchemy 2.0 Documentation

Type
Documentation

Organization
SQLAlchemy Documentation

Accessed
8 June 2026

URL

https://docs.sqlalchemy.org/en/20/orm/index.html

10. Typer

Type
Documentation

Organization
Typer

Accessed
8 June 2026

URL
https://typer.tiangolo.com/

11. Welcome to the reStructuredText Primer!

Type
Documentation

Organization
reStructuredText Primer 0.0.1 Documentation

Accessed
8 June 2026

URL
https://djdeangelis.github.io/reST-primer/index.html

12. 100% Test Coverage is a LIE, Here's Why…

Type
Video

Author
Sam Newman

Organization
Modern Software Engineering

Accessed
8 June 2026

URL
https://www.youtube.com/watch?v=p1xZ-Ni2t8Q

13. SQLAlchemy Crash Course - Master Databases in Python

Type
Video

Author
Tim Ruscica

Organization
NeuralNine

Date
Mar 17, 2025

Accessed
9 June 2026

URL
https://www.youtube.com/watch?v=529LYDgRTgQ&t

14. Prometheus Docs

Type
Documentation

Organization
The Linux Foundation

Accessed
9 June 2026

URL
https://prometheus.io/docs/introduction/overview/

15. Why 100% Test Coverage Isn't the Goal

Author
Daniel Mauno Pettersson

Organization
QA.tech

Date
March 4, 2026

Accessed
9 June 2026

URL
https://qa.tech/blog/why-100-test-coverage-isnt-the-goal

16. Test Coverage

Type
Article

Author
Fowler, Martin

Organization
Martin Fowler

Date
17 Apr. 2012

URL
https://martinfowler.com/bliki/TestCoverage.html

17. Coverage Is Not Strongly Correlated with Test Suite Effectiveness

Type
Conference Paper

Author
Inozemtseva, Laura; Holmes, Reid

Organization
ICSE 2014 / ACM

Date
2014

URL
https://dl.acm.org/doi/10.1145/2568225.2568271

18. Code Coverage and Postrelease Defects: A Large-Scale Study on Open Source Projects

Type
Journal Article

Author
Kochhar, Pavneet Singh; Lo, David; Lawall, Julia; Nagappan, Nachiappan

Organization
IEEE Transactions on Reliability

Date
2017

URL
https://ink.library.smu.edu.sg/sis_research/3838/

19. A Brief Survey on Oracle-based Test Adequacy Metrics

Type
Research Survey

Author
Hossain, Soneya Binta; Dwyer, Matthew B.

Organization
arXiv

Date
2022

URL
https://arxiv.org/abs/2212.06118

20. Four Common Types of Code Coverage

Type
Technical Article

Organization
web.dev

Date
2024

URL
https://web.dev/articles/ta-code-coverage

21. Getting 100% Code Coverage Doesn't Eliminate Bugs

Type
Technical Article

Author
Kapelonis, Kostis

Organization
Codepipes

Date
2017

URL
https://blog.codepipes.com/testing/code-coverage.html

3.3.7 Compact Works Cited

1. Goodger, David. *reStructuredText Markup Specification*. Docutils. 2026. https://docutils.sourceforge.io/docs/ref/rst/restructuredtext.html
2. OneUptime | One Complete Observability Platform. *How to Set Up a Complete Observability Stack with OpenTelemetry*. 6 Feb. 2026. https://oneuptime.com/blog/post/2026-02-06-complete-observability-stack-opentelemetry-open-source/view
3. Sheppard, Kevin. *Restructuredtext Primer - Material for Sphinx*. Accessed 8 June 2026. https://bashtage.github.io/sphinx-material/basics.html
4. OpenTelemetry. *OpenTelemetry*. Accessed 8 June 2026. https://opentelemetry.io/
5. OpenTelemetry - CNCF. *open-telemetry/opentelemetry-proto*. 22 May 2019. Accessed 8 June 2026. https://github.com/open-telemetry/opentelemetry-proto
6. OpenTelemetry. *Python OpenTelemetry*. Accessed 8 June 2026. https://opentelemetry.io/docs/languages/python/

7. Sphinx Documentation. *reStructuredText Primer - Sphinx Documentation*. Accessed 8 June 2026. https://www.sphinx-doc.org/en/master/usage/restructuredtext/basics.html
8. SQLAlchemy. *SQLAlchemy*. Accessed 8 June 2026. https://www.sqlalchemy.org
9. SQLAlchemy Documentation. *SQLAlchemy ORM - SQLAlchemy 2.0 Documentation*. Accessed 8 June 2026. https://docs.sqlalchemy.org/en/20/orm/index.html
10. Typer. *Typer*. Accessed 8 June 2026. https://typer.tiangolo.com/
11. reStructuredText Primer 0.0.1 Documentation. *Welcome to the reStructuredText Primer!*. Accessed 8 June 2026. https://djdeangelis.github.io/reST-primer/index.html
12. Sam Newman. *100% Test Coverage is a LIE, Here's Why. . .* . Modern Software Engineering. Accessed 8 June 2026. https://www.youtube.com/watch?v=p1xZ-Ni2t8Q
13. Tim Ruscica. *SQLAlchemy Crash Course - Master Databases in Python*. NeuralNine. Mar 17, 2025. Accessed 9 June 2026. https://www.youtube.com/watch?v=529LYDgRTgQ&t
14. The Linux Foundation. *Prometheus Docs*. Accessed 9 June 2026. https://prometheus.io/docs/introduction/overview/
15. Daniel Mauno Pettersson. *Why 100% Test Coverage Isn't the Goal*. QA.tech. March 4, 2026. Accessed 9 June 2026. https://qa.tech/blog/why-100-test-coverage-isnt-the-goal
16. Fowler, Martin. *Test Coverage*. Martin Fowler. 17 Apr. 2012. https://martinfowler.com/bliki/TestCoverage.html
17. Inozemtseva, Laura; Holmes, Reid. *Coverage Is Not Strongly Correlated with Test Suite Effectiveness*. ICSE 2014 / ACM. 2014. https://dl.acm.org/doi/10.1145/2568225.2568271
18. Kochhar, Pavneet Singh; Lo, David; Lawall, Julia; Nagappan, Nachiappan. *Code Coverage and Postrelease Defects: A Large-Scale Study on Open Source Projects*. IEEE Transactions on Reliability. 2017. https://ink.library.smu.edu.sg/sis_research/3838/
19. Hossain, Soneya Binta; Dwyer, Matthew B.. *A Brief Survey on Oracle-based Test Adequacy Metrics*. arXiv. 2022. https://arxiv.org/abs/2212.06118
20. web.dev. *Four Common Types of Code Coverage*. 2024. https://web.dev/articles/ta-code-coverage
21. Kapelonis, Kostis. *Getting 100% Code Coverage Doesn't Eliminate Bugs*. Codepipes. 2017. https://blog.codepipes.com/testing/code-coverage.html

3.4 Style Guide

This style guide is for me and serves as a reminder to me.

3.4.1 Scope of Guide

Seeing as I am the sole developer of this project, I don't think it is necessary to define all stylings. That being said, I will define some style choices.

BIBLIOGRAPHY

[Dijkstra1970] Dijkstra, Edsger W. *Notes on Structured Programming*. 1970.

PYTHON MODULE INDEX

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